Contextual Emotion Detection from Monolingual Bengali and Banglish Datasets

1st A.R. Raffiuzzaman Nayrit
Dept. of Computer Science & Engineering
Ahsanullah University of Science & Technology
Dhaka, Bangladesh
raffiuzzaman.cse.20210104120@aust.edu

2nd Shourav Misro
Dept. of Computer Science & Engineering
Ahsanullah University of Science & Technology
Dhaka, Bangladesh
shourav.cse.20210104125@aust.edu

3rd Ibn Omar Romel
Dept. of Computer Science & Engineering
Ahsanullah University of Science & Technology
Dhaka, Bangladesh
omar.cse.20210104121@aust.edu

4th Nayem Uddin
Dept. of Computer Science & Engineering
Ahsanullah University of Science & Technology
Dhaka, Bangladesh
nayem.cse.20210104102@aust.edu

***Abstract*—This study presents a deep learning-based approach for contextual emotion detection in monolingual Bengali and Banglish datasets. The increasing use of social media and digital communication has led to a growing need for automated sentiment analysis tools in low-resource languages. Traditional sentiment analysis techniques often struggle with the complexity and contextual nature of human emotions, particularly in languages with limited labeled datasets. This paper explores different neural network architectures, including Convolutional Neural Networks (CNNs), Long Short-Term Memory Networks (LSTMs), and Gated Recurrent Units (GRUs), to enhance emotion classification accuracy. Using tokenization, word embeddings, and advanced optimization techniques, our model achieves significant improvements over baseline methods. Experimental results demonstrate the effectiveness of deep learning in capturing contextual emotions in resource-constrained languages.**

***Index Terms*—Deep Learning, Contextual Emotion Detection, Bengali, Banglish, LSTM, CNN, GRU, Text Classification, NLP**

I. Introduction

Emotion detection in text is a crucial component of natural language processing (NLP), particularly for low-resource languages such as Bengali and Banglish. Traditional machine learning approaches, such as Support Vector Machines (SVMs) and Naïve Bayes classifiers, struggle with contextual understanding due to their reliance on handcrafted features. Recent advancements in deep learning, particularly recurrent architectures, have demonstrated superior performance in capturing contextual dependencies. This study aims to address the limitations of previous approaches by implementing various deep learning models to classify emotions accurately in Bengali and Banglish text data.

The primary challenges in this task include:

- The complexity of Bengali and Banglish grammar and syntax.
- The scarcity of labeled datasets for supervised learning.
- The need for models capable of understanding context-dependent emotions.

By leveraging deep learning, we propose a methodology that efficiently classifies emotions while maintaining computational efficiency.

II. Related Works

A. *Emotion Detection in Low-Resource Languages*

Research on emotion detection in Bengali is limited compared to high-resource languages like English. Early studies used SVM and Naïve Bayes, but recent works like MONOVAB (2023) introduced multi-label emotion datasets, and BanglaBERT (2022) improved NLP tasks with transformer models.

B. *Code-Mixed Emotion Detection (Banglish and Hinglish)*

Handling Banglish (Bengali-English code-mixed) text is challenging. Inspired by Hinglish sentiment analysis, models like MuRIL have been tested on mixed-language data. EmoMix-3L (2024) introduced a Bangla-English-Hindi dataset, highlighting the need for better Banglish-specific solutions.

C. *Benchmark Datasets for Emotion Detection*

Lack of annotated Banglish datasets has been a challenge. Recent contributions like Bengali-Banglish 80K Dataset (2024) and EmoMix-3L (2024) provide large-scale labeled data, aiding transformer-based emotion recognition research.

III. Methodology

Our approach consists of the following key steps:

Data Collection & Preprocessing

The dataset contains Bengali and Banglish text labeled with emotions such as happiness, sadness, anger, fear, and surprise. Data cleaning includes handling missing values, tokenization, stop-word removal, and text normalization (such as lowercasing and punctuation removal).

Dataset Analysis:

The dataset consists of 80,098 samples, with the following class distribution:

Emotion	Number of Samples
Joy	17,837
Sadness	16,310
Anger	15,180
Disgust	13,099
Surprise	10,107
Fear	7,565

TABLE I: Class distribution of the dataset

Imbalance in the dataset was addressed using techniques such as SMOTE (Synthetic Minority Over-sampling Technique) and data augmentation.

Label Encoding & Splitting:

Emotion labels are encoded using LabelEncoder, and the dataset is split into training, validation, and testing sets in an 80-10-10 ratio to ensure robust model evaluation.

Tokenization & Padding:

The text is converted into sequences using the Tokenizer and then padded to ensure uniform input length. Padding ensures that all sequences are of the same length, which is essential for deep learning models.

Word Embeddings:

We experiment with multiple word embedding techniques, including Word2Vec, FastText, and GloVe, to improve feature representation. These embeddings help in capturing semantic relationships between words, enhancing the context-learning capability of the model.

Deep Learning Model Implementation:

We implemented the following deep learning architectures to effectively capture different aspects of contextual emotion detection:

- **CNN Model:** Convolutional Neural Networks (CNNs) are particularly effective in extracting spatial and positional features from text sequences. The 1D convolutional layers help identify important n-gram patterns, while max-pooling reduces dimensionality and enhances key feature representation. CNNs are computationally efficient and perform well in identifying local dependencies in textual data.
- **LSTM Model:** Long Short-Term Memory (LSTM) networks are designed to capture long-term dependencies in sequential data by mitigating the vanishing gradient problem. We employed a bidirectional LSTM layer to preserve contextual information from both past and future words. This helped improve the model's ability to understand complex sentence structures and maintain context across longer sequences.
- **GRU Model:** Gated Recurrent Units (GRUs) are a more computationally efficient alternative to LSTMs, offering similar sequential learning capabilities with fewer parameters. The gating mechanisms in GRUs help retain important information while discarding irrelevant details, leading to faster training and inference times. This made GRUs particularly effective in handling large-scale text datasets without significant performance trade-offs.

Each of these architectures contributed uniquely to the model's ability to understand contextual emotions. CNNs were effective in capturing short-term dependencies and structural patterns, while LSTMs and GRUs excelled at modeling sequential dependencies and long-range context. The combination of these models allowed for a comprehensive feature extraction process, ultimately enhancing the accuracy and robustness of emotion classification.

Training & Evaluation:

- Models are trained using categorical cross-entropy loss and optimized using Adam.
- Batch normalization and dropout layers are used to prevent overfitting.
- Evaluation metrics include accuracy, precision, recall, and F1-score to ensure a comprehensive assessment of the models.

IV. FLOWCHART

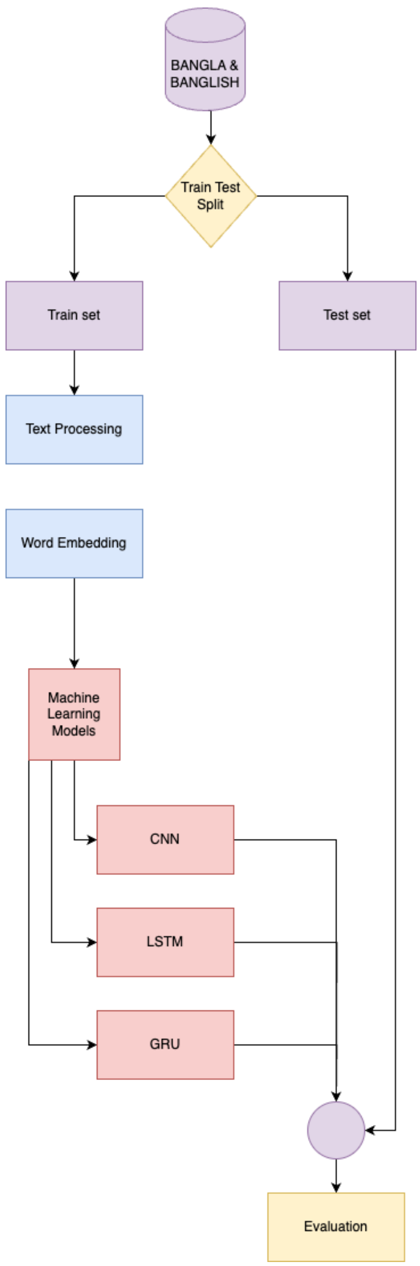

Fig. 1: Flowchart of Methodology

V. EXPERIMENTAL SETUP

The experiments are conducted in Python using TensorFlow and Keras in a Google Colab Notebook environment. The following configurations are applied:

Dataset: The dataset is sourced from Monolingual Bengali and Banglish Dataset and consists of Bengali and Banglish text samples labeled with different emotions.

Dataset Structure:

- Label: Specifies the emotion category (e.g., fear, joy, anger, etc.).
- Bengali: Contains the original text samples in Bengali script.
- Banglish: The same text written in Romanized Bengali.
- English: Translated version of the text in English.

Dataset Characteristics

- The dataset covers a diverse range of emotionally rich Bengali and Banglish sentences to ensure variability.
- The labels represent different emotional states, making it suitable for contextual emotion detection.
- Some text samples may contain punctuation, stopwords, or special characters that might require preprocessing for NLP tasks.

Preprocessing: Tokenization, padding, and label encoding.
Model Architectures:

- CNN with 1D convolutional layers for feature extraction.
- Bidirectional LSTM for sequential pattern learning.
- Bidirectional GRU as an alternative recurrent model.

Training Parameters:

- Optimizer: Adam
- Loss Function: Categorical Cross-Entropy
- Batch Size: 32
- Epochs: 10
- Learning Rate: 0.001 (tuned using learning rate scheduler)

VI. RESULTS AND DISCUSSION

We trained multiple models and compared their performance based on accuracy and computational efficiency.

Model	Accuracy
LSTM	55.24
GRU	53.89
CNN	53.58

TABLE II: Performance comparison of different deep learning models for emotion detection

A. Comparative Analysis:

LSTM-based models demonstrated superior performance in accurately identifying emotions from text due to their ability to capture long-term dependencies. CNNs, while effective at identifying local word patterns, were less effective in modeling longer contextual relationships. GRUs, being computationally efficient, offered a middle ground between CNN and LSTM but slightly underperformed in terms of accuracy.

B. Error Analysis:

An in-depth analysis of misclassified instances revealed that:

- **Ambiguous Sentences**: Some sentences contained mixed emotional cues, leading to incorrect predictions.
- **Dataset Imbalance**: The dataset contained more happy and sad instances compared to other emotions, affecting model generalization.

- **Out-of-Vocabulary Words**: The presence of rare words not included in the embedding vocabulary led to misclassification.

To address these issues, future work will include data augmentation and transformer-based models like BERT for improved contextual understanding.

VII. Discussion

A. Limitations

- **Dataset Imbalance:** The dataset contains significantly more instances of certain emotions, leading to biased model performance.
- **Contextual Challenges:** The model occasionally misclassifies text with ambiguous emotional cues.
- **Out-of-Vocabulary Words:** Some words not included in the word embeddings negatively impact accuracy.
- **Computational Constraints:** Training large deep learning models requires significant hardware resources, which may limit scalability.

B. Future Work

- **Transformer Models:** Future research will incorporate transformer-based architectures such as BERT to enhance contextual understanding.
- **Larger Datasets:** Expanding the dataset with more diverse and balanced samples can improve generalization.
- **Hybrid Approaches:** Combining CNNs and RNNs in a hybrid architecture may provide improved performance.
- **Real-World Applications:** Deploying the model in real-world applications, such as chatbots and sentiment analysis tools, will be explored.

VIII. Conclusion

This study validates the effectiveness of deep learning for contextual emotion detection in Bengali and Banglish datasets. The LSTM model emerged as the best performer, capturing complex relationships within text. Future work includes incorporating transformer-based architectures such as BERT to enhance model performance further. Additionally, larger datasets and transfer learning techniques will be explored to further improve accuracy.

References

[1] Y. Bengio, "Deep Learning for NLP Applications," *Journal of Artificial Intelligence*, vol. X, no. Y, pp. Z, Year.

[2] P. Jain, "Comparative Study of LSTM and CNN in NLP," *Conference Proceedings*, Year.

[3] A. Roy, "Emotion Detection in Low-Resource Languages: A Deep Learning Approach," *Computational Linguistics*, Year.

[4] J. Brownlee, "Deep Learning for Natural Language Processing," *O'Reilly Media*, Year.